

ARSHDEEP SINGH

✉ arsh9bl998@gmail.com 🌐 hiesenbug.dev 📄 arshdeep54 🐙 Arshdeep54

SUMMARY

Software engineer focused on backend infrastructure, observability, and systems programming. Built production backend services and asynchronous processing infrastructure, and contributed to open-source projects.

EDUCATION

Bachelor of Technology - Indian Institute of Technology, Roorkee • Roorkee **2023 - 2027**

SKILLS

Languages: Rust, C, C++, Python, JavaScript, TypeScript, Bash

Frameworks: Axum, Ratatui, BullMQ, gRPC, REST

Tools: Linux, Docker, AWS, Kubernetes, Git, PostgreSQL, Redis, Qdrant, SigNoz, Sentry, PostHog

EXPERIENCE

Software Engineer, Tal Boss (by Grapevine)  **May 2026 – August 2026, Bangalore, India**

- Worked on backend services powering job discovery and candidate matching, including candidate and job profile indexing in Qdrant and database schema improvements; experimented with Tinker Labs models for matching optimization.
- Built asynchronous background processing with BullMQ, implementing worker handlers for long-running jobs and backend workflows.
- Integrated observability with SigNoz and Sentry for debugging backend services and worked with GetStream for event handling and messaging.
- Evaluated migrating cloud storage from Azure Blob Storage to AWS S3 for cost optimization.
- Contributed to candidate onboarding infrastructure including LinkedIn authentication, profile extraction, and verification.

Contributor, Protocol Labs Dev Guild  **April 2025 - October 2025, Remote**

- Contributed to open-source projects in LabOS as part of PLDG Cohort 3.
- Built Event Harvester, an event aggregator for Luma and Cryptonomad with data cleaning and Airtable integration.
- Built Ask Racha, a docs Q&A tool that embedded parsed docs and repos into Qdrant for low-latency retrieval behind a lightweight API.

LFX Mentee, WasmEdge  **September 2025 - November 2025, Remote**

- Implemented cross-platform `wasmedgeup` installer features: version management, removal, and plugin management.
- Added system and hardware detection for runtime configuration across CPU features and CUDA, ROCm, and OpenCL backends.
- Added checksum verification, unit tests, documentation, and CI across Linux, macOS, and Windows.
- Presented the work in a WasmEdge community meeting.

PROJECTS

Volt Lab  | **Python, JavaScript, Gymnasium** **October 2026 - Present**

- Built a constrained microgrid RL environment with 15-minute energy dynamics, joint battery and EV charging actions, variable solar generation, tariffs, and outages; verified JavaScript–Python parity across complete simulation weeks.
- Benchmarked Q-learning and Stable-Baselines3 DQN across three training seeds and 30 held-out synthetic weeks with approximately matched transition budgets; continuous-observation DQN achieved a 4.7% lower mean mixed-weather objective than a tariff-aware heuristic.
- Analyzed policy failures through reward ablations, outage stress tests, and cost decomposition; added learning curves against interactions and training time, reproducible model artifacts, and policy imports with independently recomputed evaluation scores.

VortexDB  | **Rust** **October 2025 - Present**

- Built a lightweight vector database in Rust with pluggable Flat, KD-Tree, and HNSW indexing and Euclidean, Manhattan, Hamming, and Cosine distance metrics.
- Designed a pluggable persistence layer supporting in-memory storage and RocksDB through a common `StorageEngine` interface.
- Exposed the database through Axum HTTP and gRPC APIs and built a Ratatui terminal UI for interacting with the system.

agentq  | **Rust** **September 2026 - Present**

- Built an embeddable job queue for agent tool calls: idempotency keys cache outputs, so retries return cached results and prevent duplicate writes and charges.
- Added priority lanes with independent concurrency limits and backpressure against unbounded queue growth; embedded directly in the binary with failure isolation and cancel-safe semantics, published on crates.io with no external service or Redis dependency.

OPEN SOURCE

- Contributed to **GreptimeDB**, an open-source time-series database for observability, and the PostgreSQL tooling projects **pgmoneta**, **pgexporter**, and **pgagroal**.

RECOMMENDATIONS

Jesper Pedersen

Google Summer of Code Administrator • PostgreSQL • jesperpedersen.db@gmail.com